

## TERRELL, TX, USA

WMO: 722489

Lat: 32.710N

Lon: 96.267W

Elev: 475

StdP: 14.45

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 53911

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 23.9       | 27.5 | 9.6                           | 9.1 | 32.2 | 14.2 | 11.5 | 36.3 | 24.4                  | 53.3 | 21.7 | 56.2 | 6.3                   | 0    | 0.462 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 8             | 20.7                      | 100.4           | 74.4 | 98.6 | 74.8 | 96.3 | 74.7 | 78.4                | 90.8 | 77.7 | 90.1 | 77.1 | 89.2 | 8.2                     | 160  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 75.3                            | 134.9 | 82.4 | 74.6 | 131.8 | 82.0 | 73.6 | 127.3 | 81.3 | 42.1          | 91.0 | 41.4 | 90.2 | 40.7 | 89.1 |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |    |    |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|----|----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |    |    |
| 21.0              | 19.0 | 17.3 | 17.6                       | 104.2 | 3.8                | 3.2 | 14.9                                               | 106.5 | 12.7       | 108.3 | 10.5       | 110.1 | 7.8        | 112.4 |    |    |
|                   |      |      | 15.7                       | 79.8  | 3.5                | 1.2 | 13.1                                               | 80.6  | 11.1       | 81.3  | 9.1        | 82.0  | 6.6        | 82.8  |    |    |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 66.5   | 46.6  | 50.3  | 58.4  | 65.8  | 73.9  | 81.4  | 84.6  | 85.2  | 78.5  | 67.6  | 56.9  | 48.2 |
|                                                                               | DBStd       | 15.69  | 9.84  | 10.37 | 9.07  | 7.37  | 6.21  | 3.96  | 3.52  | 4.30  | 5.84  | 8.10  | 9.46  | 9.51 |
|                                                                               | HDD50       | 523    | 184   | 113   | 31    | 2     | 0     | 0     | 0     | 0     | 0     | 2     | 41    | 150  |
|                                                                               | HDD65       | 2186   | 574   | 419   | 239   | 79    | 11    | 0     | 0     | 0     | 1     | 67    | 269   | 527  |
|                                                                               | CDD50       | 6559   | 78    | 122   | 291   | 475   | 740   | 943   | 1073  | 1091  | 855   | 547   | 249   | 95   |
|                                                                               | CDD65       | 2747   | 3     | 8     | 34    | 102   | 286   | 493   | 608   | 626   | 406   | 147   | 28    | 6    |
|                                                                               | CDH74       | 30833  | 15    | 45    | 197   | 735   | 2585  | 5548  | 7713  | 8141  | 4356  | 1314  | 159   | 25   |
|                                                                               | CDH80       | 15077  | 0     | 3     | 20    | 140   | 919   | 2673  | 4196  | 4589  | 2085  | 432   | 19    | 1    |
| Wind                                                                          | WSAvg       | 8.0    | 8.2   | 8.8   | 9.3   | 9.4   | 8.7   | 8.0   | 7.1   | 6.5   | 6.5   | 7.2   | 7.9   | 8.0  |
| Precipitation                                                                 | PrecAvg     | 39.1   | 3.0   | 2.9   | 3.8   | 3.4   | 4.8   | 3.4   | 1.8   | 2.3   | 2.9   | 4.6   | 3.5   | 3.4  |
|                                                                               | PrecMax     | 54.8   | 8.4   | 6.7   | 8.3   | 8.5   | 13.2  | 10.2  | 4.7   | 6.5   | 9.3   | 13.1  | 9.5   | 8.7  |
|                                                                               | PrecMin     | 23.1   | 0.3   | 0.2   | 0.4   | 0.6   | 0.7   | 0.6   | 0.0   | 0.0   | 0.2   | 0.3   | 0.4   | 0.3  |
|                                                                               | PrecStd     | 9.0    | 1.8   | 1.7   | 2.0   | 2.0   | 2.8   | 2.4   | 1.4   | 1.7   | 1.9   | 3.5   | 2.2   | 2.2  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 76.1  | 79.1  | 82.6  | 87.9  | 94.1  | 99.4  | 102.1 | 104.3 | 100.5 | 91.5  | 82.5  | 77.3 |
|                                                                               |             | MCWB   | 60.7  | 63.6  | 65.0  | 70.1  | 73.2  | 74.2  | 74.0  | 72.8  | 71.8  | 72.1  | 67.1  | 64.1 |
|                                                                               | 2%          | DB     | 71.9  | 74.6  | 79.1  | 83.6  | 90.4  | 96.4  | 99.5  | 101.5 | 96.1  | 88.4  | 78.6  | 72.4 |
|                                                                               |             | MCWB   | 60.7  | 61.2  | 65.0  | 69.0  | 73.4  | 74.4  | 75.0  | 74.3  | 73.4  | 70.8  | 65.9  | 62.5 |
|                                                                               | 5%          | DB     | 67.8  | 71.0  | 75.8  | 81.1  | 87.9  | 93.7  | 97.6  | 99.3  | 93.1  | 85.0  | 75.0  | 68.5 |
|                                                                               |             | MCWB   | 58.7  | 61.2  | 63.3  | 67.9  | 72.7  | 74.5  | 75.2  | 75.1  | 73.0  | 68.7  | 64.2  | 61.0 |
|                                                                               | 10%         | DB     | 63.8  | 67.1  | 72.7  | 78.7  | 85.1  | 91.4  | 95.3  | 97.0  | 90.7  | 81.5  | 72.0  | 64.5 |
|                                                                               |             | MCWB   | 56.4  | 59.1  | 62.3  | 66.6  | 71.7  | 74.8  | 75.4  | 74.7  | 72.5  | 67.8  | 63.2  | 57.5 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 66.6  | 67.8  | 69.8  | 74.9  | 77.7  | 79.4  | 79.3  | 79.2  | 77.8  | 75.9  | 71.4  | 68.4 |
|                                                                               |             | MCDB   | 70.9  | 74.7  | 77.4  | 82.7  | 86.8  | 89.8  | 92.9  | 92.9  | 89.0  | 84.1  | 77.4  | 72.8 |
|                                                                               | 2%          | WB     | 64.4  | 65.7  | 67.7  | 72.3  | 76.1  | 78.1  | 78.3  | 78.2  | 76.8  | 74.4  | 69.0  | 65.7 |
|                                                                               |             | MCDB   | 68.3  | 71.3  | 74.1  | 79.9  | 85.0  | 88.8  | 91.6  | 91.7  | 88.1  | 82.8  | 74.0  | 70.5 |
|                                                                               | 5%          | WB     | 61.6  | 63.8  | 66.3  | 70.6  | 75.0  | 77.3  | 77.6  | 77.4  | 75.9  | 73.1  | 67.0  | 62.8 |
|                                                                               |             | MCDB   | 66.7  | 67.9  | 72.9  | 77.5  | 83.9  | 88.0  | 90.7  | 90.6  | 86.7  | 80.6  | 72.8  | 67.0 |
|                                                                               | 10%         | WB     | 56.7  | 60.7  | 64.4  | 69.0  | 73.8  | 76.4  | 77.0  | 76.7  | 75.0  | 71.3  | 64.8  | 59.1 |
|                                                                               |             | MCDB   | 62.1  | 66.6  | 71.0  | 75.4  | 82.6  | 86.9  | 89.6  | 89.6  | 85.3  | 77.7  | 70.6  | 64.1 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 20.5  | 19.6  | 20.2  | 20.4  | 19.1  | 18.4  | 20.0  | 20.7  | 20.9  | 21.6  | 20.4  | 19.9 |
|                                                                               |             | MCDBR  | 25.7  | 23.9  | 21.8  | 21.6  | 21.1  | 21.0  | 22.5  | 23.9  | 23.0  | 23.9  | 22.4  | 23.7 |
|                                                                               |             | MCWBR  | 16.3  | 15.1  | 11.9  | 11.0  | 8.3   | 5.9   | 5.4   | 5.3   | 6.9   | 9.9   | 12.4  | 15.8 |
|                                                                               | 5% WB       | MCDBR  | 20.9  | 20.0  | 18.3  | 17.4  | 17.7  | 17.3  | 19.3  | 20.3  | 19.0  | 18.3  | 17.7  | 20.1 |
|                                                                               |             | MCWBR  | 16.3  | 15.1  | 12.2  | 11.0  | 8.9   | 6.2   | 5.9   | 5.6   | 6.4   | 9.6   | 12.7  | 16.2 |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.317  | 0.334 | 0.359 | 0.396 | 0.428 | 0.440 | 0.444 | 0.446 | 0.400 | 0.359 | 0.340 | 0.322 |      |
|                                                                               | taud        | 2.496  | 2.458 | 2.406 | 2.323 | 2.291 | 2.289 | 2.317 | 2.298 | 2.376 | 2.474 | 2.470 | 2.486 |      |
|                                                                               | Ebn at Noon | 287    | 292   | 292   | 284   | 274   | 269   | 267   | 266   | 274   | 279   | 275   | 278   |      |
|                                                                               | Edh at Noon | 28     | 31    | 36    | 41    | 42    | 42    | 41    | 41    | 36    | 31    | 28    | 26    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 890    | 1025  | 1382  | 1692  | 1828  | 2050  | 2094  | 1933  | 1626  | 1281  | 949   | 786   |      |
|                                                                               | RadStd      | 86     | 163   | 114   | 118   | 150   | 192   | 143   | 131   | 141   | 188   | 121   | 102   |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (a)  | (b)                     | (c)   | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (44 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page